

SourceGuard-Gait: Claim-Bounded Invariance for Heterogeneous Public-Source Abnormal-Gait Screening

Anonymous submission

Abstract

Public gait resources are increasingly assembled from unpaired multi-modal signals, including wearable IMU recordings, RGB videos, silhouette masks, human skeletons, and body meshes. Their core challenge is support admissibility: source suppression is empirically meaningful only in label-modality strata where comparable cross-source support exists. In single-source strata, forcing invariance can erase abnormal-gait evidence and overstate robustness. We study this setting as a claim-bounded invariance protocol after normalizing inputs into IMU trajectories and multi-view gait sequences. SourceGuard-Gait uses a train-split source-modality-label support matrix to decide when source suppression and distribution alignment are admissible, then audits frozen embeddings for residual source recoverability. Extensive experiments show that SourceGuard-Gait achieves consistent improvements in screening performance, while effectively mitigating residual source information.

Introduction

Gait is an informative behavioral signal for neurological and mobility assessment, but public abnormal-gait data rarely form a clean supervised benchmark. The sources used here span wearable Parkinsonian recordings from WearGait-PD (Kontson et al. 2024; Anderson et al. 2026), biometric visual gait sequences from CASIA-B (Yu, Tan, and Tan 2006; Institute of Automation, Chinese Academy of Sciences 2016), video-based GAIT-IT (Instituto de Telecomunicacoes n.d.b) and GAIT-IST (Instituto de Telecomunicacoes n.d.a) clinical resources, public clinical gait videos from GAVD (Ranjan et al. 2025a,b), functional adult videos from Health&Gait (Zafra-Palma et al. 2025), multi-site Parkinsonian meshes from CARE-PD (Adeli et al. 2025), and depression-risk visual gait sequences from D-Gait (Liu et al. 2024). In this paper, wearable records enter the model as IMU time series, while video, silhouette, skeleton, and mesh resources are converted into multi-view gait sequences. The target is a public-source abnormal-gait screening protocol; disease-diagnostic endpoints require separate clinical validation. The sources still differ in population, acquisition protocol, label

vocabulary, and preprocessing pipeline, so pooled training must account for source-specific support and label structure.

A pooled binary classifier can exploit these differences, for example by associating a screen-negative reference repository or a disease-specific wearable protocol with the label. This is a familiar shortcut-learning failure mode (Geirhos et al. 2020; DeGrave, Janizek, and Lee 2021; Ong Ly et al. 2024). Source suppression is a natural response to this failure mode because it attempts to reduce the amount of source information carried by the representation. The difficulty is that suppression itself requires evidence: a model can learn to remove source-specific nuisance variation only when multiple sources instantiate the same label-modality condition.

In a label-modality-acquisition stratum represented by a single source, a conditional source critic lacks the cross-source variation needed to separate removable provenance from disease-relevant movement evidence. An unqualified source-suppression objective can then penalize the very movement pattern that identifies pathology. SourceGuard-Gait uses this support-admissibility constraint as its organizing principle. The support matrix determines which strata receive source-oriented regularization, which strata are reported as support gaps, and how each robustness claim is interpreted.

These observations lead to the central idea of the paper: heterogeneous gait learning needs a support-aware robustness contract that ties each source-suppression operation to the evidence available for that stratum and audits the remaining source signal. SourceGuard-Gait implements this contract as claim-bounded invariance. Our main contributions can be summarized as follows:

First, we formulate heterogeneous public gait learning as a support-aware screening problem and give a concise source-suppression gate. The formulation separates movement evidence used for prediction from source, subject, acquisition, and condition-family variables used for splitting, routing, reporting, and residual-source evaluation.

Second, we introduce SourceGuard-Gait as a claim-bounded invariance protocol centered on the support matrix. The matrix gates adversarial source suppression and distribution alignment, keeps unsupported strata in screening learning, and converts missing cross-source evidence into explicit support-gap reports.

Third, we evaluate the method with claim-matched checks:

This is an anonymized submission for review purposes only. Distribution, citation, or public sharing of this manuscript is strictly prohibited. Copyright and publication details will appear in the final version if accepted.

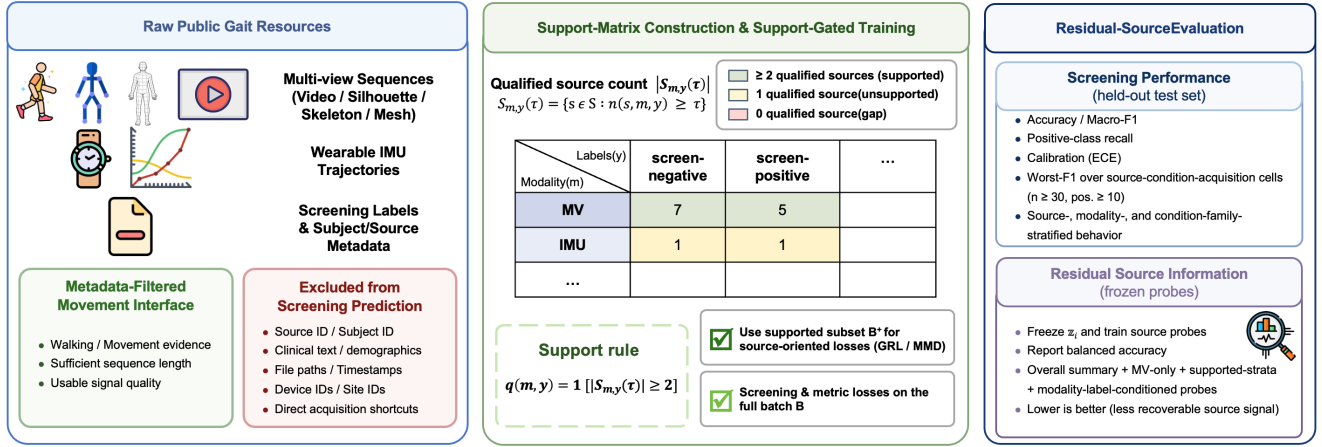


Figure 1: Overview of SourceGuard-Gait. Heterogeneous public gait sources are first passed through a metadata-filtered movement interface, summarized into a source-modality-label support matrix, encoded by modality-specific branches, and evaluated through screening, calibration, worst-stratum, and frozen source-probe readouts. Conditional source suppression is interpreted only where comparable support exists.

held-out screening, source-stratified behavior, calibration, worst-stratum performance, frozen source probes, multi-seed stability, leave-one-source-out transfer, and source-only, modality-only, label-permutation, within-source, and matched-source controls.

Related Work

Gait Representation Learning

Template-based methods such as Gait Energy Images compress sequences for identity recognition (Han and Bhanu 2006). GaitSet models gait as an unordered silhouette set and aggregates frame-level features for cross-view recognition (Chao et al. 2019). GaitPart captures fine-grained part-level cues (Fan et al. 2020), while QAGait highlights the importance of silhouette quality under in-the-wild conditions (Wang et al. 2024). In-the-wild, multimodal, skeleton-based, and micro-Doppler studies reveal variation in view, representation, pose estimation, and sensor domain (Guo et al. 2025; Cui and Kang 2023; Teepe et al. 2022; Yang et al. 2022). Abnormal-gait research further explores self-supervised detection, cross-subject and cross-modal transfer, multi-source adaptation, and privacy-preserving skeleton-based screening (Duan, Wan, and Zhao 2024; Gu et al. 2021; Guo, Gu, and Yang 2021; Naz et al. 2025). These methods strengthen gait evidence, but clinical screening with cross-source label entanglement also requires evaluation under source, modality, and label support constraints.

Clinical Gait Benchmarks

Recent clinical gait resources increasingly underscore the need for cross-source evaluation. CARE-PD targets multi-site Parkinsonian gait assessment, while WearGait-PD provides wearable recordings from patients and controls (Adeli et al. 2025; Kontson et al. 2024; Anderson et al. 2026). Video-based resources such as GAVD and skeleton gait-energy studies show that public videos capture clinically

relevant movement patterns (Ranjan et al. 2025a,b; Loureiro and Correia 2020). Multimodal studies integrate skeletons, plantar pressure, wearables, and sensor fusion under controlled or free-living conditions (Jun et al. 2021; Celik et al. 2022). Health&Gait and D-Gait provide controlled visual sequences with functional or risk labels (Zafra-Palma et al. 2025; Liu et al. 2024), while IAPR-indexed repositories add diversity in scale, view, clothing, occlusion, age, and sensing (IAPR Technical Committee 4 n.d.). Many of these repositories provide identity or covariate labels rather than clinical condition labels. In this study, they are therefore used only as reference-negative datasets or controls and are reported separately from clinical datasets.

Domain Generalization and Support-Admissible Invariance

Domain-adversarial learning promotes feature invariance via gradient reversal (Ganin et al. 2016), while CDAN conditions alignment on classifier information (Long et al. 2018). GroupDRO targets worst-group performance (Sagawa et al. 2020), and IRM and REX encourage cross-environment invariance (Arjovsky et al. 2019; Krueger et al. 2021). Benchmark studies highlight the fragility of domain-generalization gains and the need for realistic, metadata-structured shifts (Gulrajani and Lopez-Paz 2021; Koh et al. 2021). Other approaches include style-shift regularization (Zhou et al. 2021, 2023) and feature alignment through MMD or covariance matching, as in Deep CORAL (Gretton et al. 2012; Sun and Saenko 2016). SourceGuard-Gait adapts these ideas through a support-admissibility test that interprets each source-suppression term according to observed source-modality-label support.

Problem Formulation

Public-source gait learning is constrained not only by data scarcity, but also by the support structure of the available

evidence. The sources differ in sensing modality, label convention, acquisition protocol, subject population, and pre-processing pipeline. The problem is to learn a movement-based screen-negative/screen-positive predictor while making every source-suppression claim conditional on comparable cross-source evidence.

Task Definition

Each observed record is represented as

$$o_i = (x_i, y_i, s_i, m_i, u_i, a_i, c_i),$$

where $x_i \in \mathcal{X}_{m_i}$ is movement evidence in the observation space of modality m_i , $y_i \in \{0, 1\}$ is the binary screening label, $s_i \in \mathcal{S}$ is the source identifier, u_i is the subject key used for splitting, a_i records acquisition context when it is available, and c_i is the retained condition-family tag used for stratified reporting. The modality-indexed space $\mathcal{X}_{m_i}$ distinguishes multi-view gait sequences from wearable time series; the modalities are collected from unpaired subjects and trials.

The model receives x_i and m_i as predictive input and outputs

$$\hat{p}_i = f_\theta(x_i, m_i) \in [0, 1],$$

the estimated probability that the sample belongs to the pathological screening class. Source, subject, and acquisition variables are reserved for split construction, support estimation, routing, and residual-source evaluation. The screening predictor receives movement evidence and modality only. The learning target is high screening utility on subject-disjoint held-out records, measured by pathological F1, recall, calibration, and worst-stratum behavior, while reducing recoverable source information only in strata where source suppression is support-admissible.

The central obstacle is source-label entanglement. Let S , M , A , and Y denote source, modality, acquisition context, and screening label. In heterogeneous public gait data,

$$Y \not\perp (S, M, A).$$

A source may contain only controls, another may contain a disease-specific population, and another may provide a distinct sensing protocol. As a result, $P(Y \mid S, M, A)$ can be informative even when the movement evidence alone is the intended basis for prediction. This creates the usual shortcut-learning risk and a more specific ambiguity for source suppression: in a fixed modality-label stratum represented by a single source, a feature can be explained either as removable acquisition nuisance or as valid screening evidence expressed by that source. Because the two explanations induce the same observed distribution over (X, Y, S, M) , the data do not justify deciding which explanation is correct. Such strata are retained for screening and reporting, while source-suppression claims are withheld.

The minimal support gate formalizes this distinction using independent subject-key support, so derived clips, views, or rendered frames cannot inflate admissibility:

$$n_u(s, m, y) = |\{u_i : s_i = s, m_i = m, y_i = y, i \in \mathcal{D}_{\text{train}}\}|.$$

For a threshold τ , the comparable source set for a modality-label stratum is

$$\mathcal{S}_{m,y}(\tau) = \{s \in \mathcal{S} : n_u(s, m, y) \geq \tau\}.$$

The support indicator is then

$$q(m, y) = \mathbf{1}[|\mathcal{S}_{m,y}(\tau)| \geq 2].$$

When $q(m, y) = 1$, source suppression has an empirical cross-source contrast and is support-admissible for that modality-label stratum. When $q(m, y) = 0$, the same examples can still contribute to screening evaluation and source-stratified reporting, but the corresponding source-suppression loss and robustness claim are disabled. The gate is a count-based admissibility rule rather than a guarantee of clinical comparability; condition-family reporting and the conservative family-aware gate $q(m, y, c)$ are used to audit whether cross-family alignment changes the conclusion.

Claim Boundary

A useful representation in this paper preserves abnormal-gait discriminability while reducing recoverable source information in support-admissible strata. The scope is heterogeneous public-source abnormal-gait screening with unpaired modalities. Clinical diagnosis, disease-subtype classification, and patient-level paired multimodal fusion require additional evidence beyond this manifest. SourceGuard-Gait therefore reports screening utility, support validity, and residual source recoverability together, tying each robustness statement to the evidence available for the relevant stratum.

SourceGuard-Gait

SourceGuard-Gait operationalizes the formulation above as a support-gated representation-learning framework. Its pipeline is: metadata-filtered movement parsing, subject-disjoint splitting, training-split support-matrix construction, support-gated training, modality-specific encoding, shared screening embedding, and residual-source evaluation.

Overview

Using the variables defined in the problem formulation, SourceGuard-Gait maps each admitted movement sample into a shared embedding $z_i \in \mathbb{R}^d$, with $d = 256$:

$$\begin{aligned} z_i &= F_\theta(G_{m_i}(x_i)), \\ \hat{p}_i &= \sigma(w^\top z_i + b). \end{aligned}$$

Here G_{m_i} is the encoder selected by modality, F_θ is a lightweight residual projection into the shared space, and $w^\top z_i + b$ is a scalar binary screening head. The support gate controls which source-based losses and claims are active for the stratum containing sample i .

Metadata-Filtered Movement Interface

The metadata-filtered interface controls the transition from raw public records to model inputs. Multi-view sequence inputs include rendered meshes, silhouettes, GEIs, SEIs, pose templates, or frame sequences normalized to 64×64 . IMU inputs include numeric sensor trajectories after removing high-risk administrative or clinical fields. Source identifiers, subject identifiers, file paths, clinical text, demographic descriptors, event markers, contact indicators, and time-like columns are retained for grouping or probing when needed

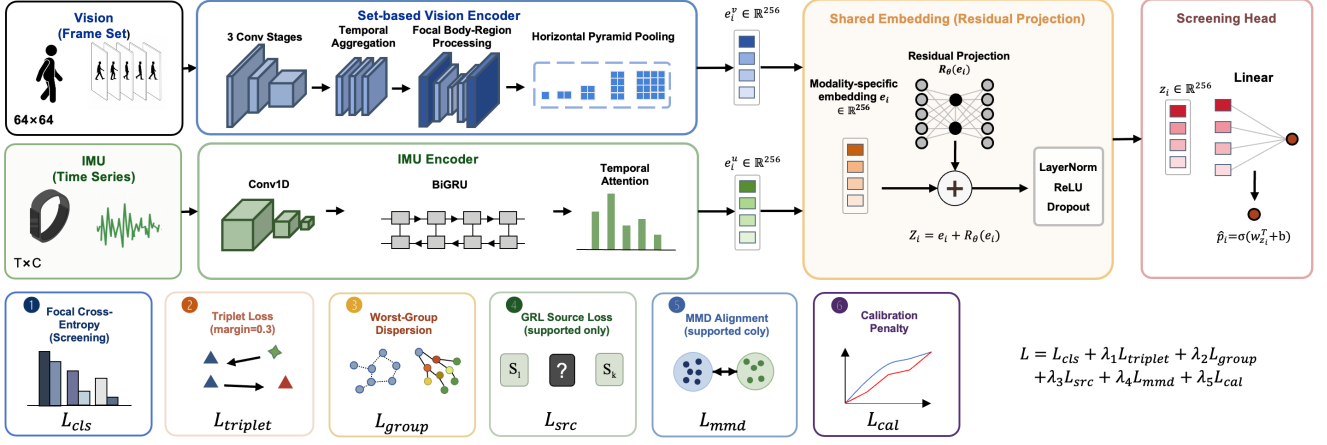


Figure 2: Training graph of SourceGuard-Gait. The metadata-filtered interface admits movement evidence, modality-specific encoders map multi-view sequence and IMU samples into a shared embedding, and the training objective combines screening, metric, group, source, alignment, and calibration terms.

and excluded from screening prediction. This filtering reduces direct metadata leakage, while source probes audit residual visual artifacts from cameras, crops, or extractors.

Support Matrix and Support-Gated Routing

The support matrix is the core routing object in SourceGuard-Gait. It is computed after the subject-disjoint split and before model fitting, using only training records. If $q(m, y) = 1$, examples in that modality-label stratum have a cross-source comparison basis, and source regularization and source alignment are allowed. If $q(m, y) = 0$, the screening loss still uses the example, but source suppression terms are skipped for that stratum and the result is reported as a support gap.

For a mini-batch B , let $q_i = q(m_i, y_i)$ and $B^+ = \{i \in B : q_i = 1\}$. Source-oriented losses are evaluated on B^+ . Screening and metric-learning losses are evaluated on the full movement batch. This routing rule turns the support matrix into a gate over the learning objective and preserves unsupported samples for screening learning and support-gap reporting. Table 1 enumerates the observed modality-label routes in the active manifest. Alternative policies that fully enable alignment, partially enable it through soft counts, restrict it by condition family, or over-align all routes are evaluated in Table 4.

The same support view also shapes mini-batch construction. Let $g = (s, y)$ be a nonempty source-label group in the training split. Each batch samples eight groups uniformly from the nonempty group set and then samples four instances with replacement inside each selected group. Thus every represented group contributes the same per-batch cap, including support-gap strata, while large screen-negative sequence corpora cannot dominate the gradient before smaller clinical strata are seen. Under $\tau = 50$, the supported multi-view routes contain 28,570 training rows after preprocessing expansion; support itself is computed from subject keys so that derived clips, templates, or rendered views cannot inflate the gate.

Stratum	Sources passing subject support	q	Src. loss	Claim
IMU-neg.	WearGait-PD only	0	off	support gap
IMU-pos.	WearGait-PD only	0	off	support gap
MV-neg.	multiple reference/clinical MV sources	1	on	count-supported
MV-pos.	multiple clinical/risk MV sources	1	on	count-supported

Table 1: Support-gated routing under $\tau = 50$. These four rows exhaust the observed modality-label routes; fully open, soft, family-aware, and over-aligned policies are evaluated as ablations in Table 4.

Modality Encoders

The architectural novelty lies in support-aware routing and residual-source testing; the encoders instantiate that framework for multi-view sequence and wearable gait inputs.

The multi-view branch follows set-based gait recognition practice (Chao et al. 2019): frames, GEI/SEI templates, pose templates, and rendered views are normalized to 64×64 , lifted by convolutional stages, aggregated over time, and pooled over horizontal body regions. Learned feature stages use 32, 64, and 128 channels after the input image has been lifted from the spatial grid; these are feature channels, not input dimensions. The wearable branch treats each record as a multichannel time series, removes metadata and event columns, standardizes each channel within sequence, clips standardized values to $[-5, 5]$, and encodes short- and long-range motion with temporal convolutions, a bidirectional recurrent layer, and attention. Both branches output a 256-dimensional vector.

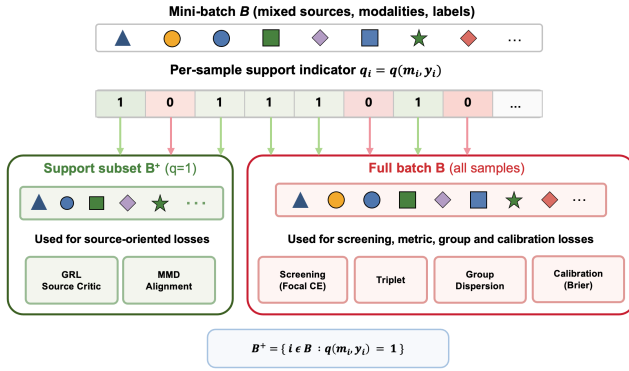


Figure 3: Support-gated mini-batch routing. The support indicator $q(m, y)$ routes comparable examples to source-oriented losses, and the full movement batch remains available for screening and metric learning.

Shared Embedding and Screening Head

The shared projection receives whichever modality is present for a sample. When a mini-batch contains both multi-view sequence and IMU examples, their encoded vectors are represented as separate batch elements in the common embedding space. The projection is residual:

$$z_i = e_i + R_\theta(e_i),$$

where e_i is the modality-specific embedding and R_θ is a two-layer MLP with LayerNorm, ReLU, and dropout. A linear head maps z_i to the screen-positive probability.

Learning Objective

The learning objective combines standard components with support-gated routing. Focal cross-entropy performs screening (Lin et al. 2017); triplet loss makes the embedding label-discriminative (Schroff, Kalenichenko, and Philbin 2015); group dispersion improves worst-group stability; a stratum-conditional GRL source loss and MMD alignment discourage source separability only where support permits; and a differentiable Brier-style calibration penalty encourages probabilistic calibration. Expected calibration error is used only as an evaluation metric.

Let $r_i = (m_i, y_i)$ be the main support route and let $\mathcal{R}^+ = \{r : |\mathcal{S}_r(\tau)| \geq 2\}$. For a mini-batch B , define $B_r = \{i \in B : r_i = r\}$, $B_{r,s} = \{i \in B_r : s_i = s\}$, $Z_{r,s} = \{z_i : i \in B_{r,s}\}$, and valid source-pair set $\mathcal{P}_r = \{(s, t) : s < t, |B_{r,s}| > 0, |B_{r,t}| > 0\}$. The source and MMD terms are

$$\mathcal{L}_{\text{src}} = \sum_{r \in \mathcal{R}^+} \frac{1}{|B_r|} \sum_{i \in B_r} \text{CE}(D_r(z_i), s_i),$$

$$\mathcal{L}_{\text{mmd}} = \sum_{r \in \mathcal{R}^+} \frac{1}{|\mathcal{P}_r|} \sum_{(s,t) \in \mathcal{P}_r} \text{MMD}_k^2(Z_{r,s}, Z_{r,t}).$$

where routes with $B_r = \emptyset$ or $\mathcal{P}_r = \emptyset$ are skipped. The source critic is a one-hidden-layer MLP and minimizes cross-entropy, while the encoder receives the reversed gradient through the GRL. MMD uses a Gaussian RBF kernel with the

Setting	Value
Embedding dimension	256
Sequence resolution	64×64
Training frames	30
Batch structure	8 source-label groups $\times$ 4 instances
Optimizer	AdamW, 5×10^{-4} learning rate
Triplet margin	0.3
Focal loss	$\gamma = 2, \alpha = (0.9, 1.1)$
Regularization weights	group 0.1, source coefficient 0.2, MMD 0.1, Brier 0.05
Decision rule	fixed threshold 0.5

Table 2: Training configuration used by SourceGuard-Gait.

bandwidth set by the route-level median heuristic. For group dispersion, G_B contains nonempty source-label-condition groups in the batch, ℓ_g is the mean screening loss in group g , and

$$\mathcal{L}_{\text{grp}} = |G_B|^{-1} \sum_{g \in G_B} (\ell_g - \bar{\ell})^2,$$

$$\mathcal{L}_{\text{cal}} = |B|^{-1} \sum_{i \in B} (\hat{p}_i - y_i)^2.$$

Triplets use anchors and positives with the same screening label and condition family when available, negatives with the opposite label, and cross-source positives only in support-admissible strata. The family-aware ablation replaces $r = (m, y)$ with $r = (m, y, c)$. The encoder is trained with

$$\mathcal{L} = \lambda_{\text{cls}} \mathcal{L}_{\text{cls}} + \lambda_{\text{tri}} \mathcal{L}_{\text{tri}} + \lambda_{\text{grp}} \mathcal{L}_{\text{grp}} + \lambda_{\text{src}} \mathcal{L}_{\text{src}} + \lambda_{\text{mmd}} \mathcal{L}_{\text{mmd}} + \lambda_{\text{cal}} \mathcal{L}_{\text{cal}}.$$

Residual Source Evaluation

The source critic used during training provides optimization pressure, while a separate frozen probe gives a stronger evaluation of residual source information. A critic may be weak, undertrained, or defeated by optimization dynamics while source cues remain accessible. SourceGuard-Gait therefore freezes the best screening encoder and trains source probes after training: The evaluation uses a linear probe and a shallow MLP. Probe accuracy is reported with chance and balanced accuracy; the main residual-source claim uses multi-view-only and modality-label-conditioned probes so recoverability is not dominated by WearGait-PD being the only IMU source.

Experiments

The experiments test whether support-gated source suppression improves abnormal-gait screening while making incomplete public-source support auditable. The questions are screening utility, support validity, and residual source information. Subject-disjoint means that a participant or parent recording key appears in only one split within each source; all derived views, clips, templates, and renders inherit that

split. Every source-suppression claim is paired with frozen-embedding source-probe evaluation.

Unless otherwise stated, SourceGuard-Gait denotes the full configuration with metadata-filtered movement inputs, subject-disjoint splitting, source-balanced group sampling, support-gated source regularization, stratum-conditional alignment, a Brier-style calibration penalty, and frozen source-probe evaluation. Learned comparisons use the same split, preprocessing interface, class balancing, threshold 0.5, seed list, and hyperparameter budget. Diagnostic controls change only the input cue or post-hoc operation named in the row. The + rows in Table 4 are cumulative ablations; the + sampler row is the pooled classifier after adding source-balanced sampling.

Data Manifest and Split

The experimental manifest is built from public sources only. Each usable record is converted into one of two admitted inputs: an IMU trajectory or a multi-view gait sequence. GAVD videos are retrieved from public links and converted into person-centric frame sequences using the released annotations; CARE-PD meshes are rendered from canonical walking views; and video or silhouette sources are normalized into the same visual sequence interface (Ranjan et al. 2025a,b; Adeli et al. 2025; Zafra-Palma et al. 2025; Liu et al. 2024). The binary label is assigned as screen-negative or screen-positive, with the original condition family retained for stratified evaluation. The split is performed within each source by subject key before any source-balanced sampling or support computation. Every derived view, trial, clip, GEI/SEI aggregate, and rendered camera inherits the same subject key as its parent recording, so derived samples cannot cross train/test boundaries.

Before splitting, we apply source-specific movement extraction: IMU records are filtered for complete walking trajectories, silhouettes and videos are checked for usable person-centric sequences, and mesh records are rendered only when canonical walking views can be produced. This process excludes 596 of 43,872 raw units, leaving 43,276 usable units for the manifest. The excluded cases are dominated by sensor gaps, incomplete silhouettes, short clips, unavailable public videos, mesh rendering errors, and tracking failures. Failure logs are retained for bias checks and are never used as model inputs.

Implementation Details

Multi-view gait sequences are normalized to 64×64 . Training samples 30 frames per view when temporal clips are available; GEI/SEI-style aggregates are repeated only to match the sequence interface. Mesh sources are rendered from fixed canonical cameras, public videos are cropped around the annotated walking subject, and wearable records are standardized within sequence. Split files are generated over source-subject units before preprocessing expansion. Support matrices and source-balanced sampling groups are computed from the training split only, while hyperparameters are selected on a source-stratified validation split. The

main hyperparameters are in Table 2; the objective is

$$\mathcal{L} = \mathcal{L}_{\text{cls}} + \mathcal{L}_{\text{tri}} + 0.1\mathcal{L}_{\text{grp}} + 0.2\mathcal{L}_{\text{src}} + 0.1\mathcal{L}_{\text{mmd}} + 0.05\mathcal{L}_{\text{cal}}.$$

The source term is written with a positive coefficient because the gradient reversal layer supplies the sign change to the encoder. The coefficient 0.2 is the base source-loss weight; the GRL multiplier λ_s starts at 1.0 and increases when validation F1 plateaus, with an upper bound of 5.0, so the effective reversed encoder pressure is $0.2\lambda_s$. The source critic is optimized first on each batch; the encoder is then updated after a second forward pass.

Metrics and Probe Protocol

The primary screening metrics are accuracy, precision, pathological recall, and F1 score, with class 1 denoting pathological gait. We also report macro-F1, PR-AUC, ten-bin expected calibration error (Guo et al. 2017), and confusion counts. Worst-F1 is the minimum positive-class F1 over pre-defined source-condition-acquisition audit cells with at least 30 held-out samples and at least 10 positives. The worst SourceGuard-Gait cell is an uncontrolled oblique-video subset of GAVD with $n = 146$, 61 negatives, 85 positives, $TP/FP/FN = 63/26/22$, recall 74.1, and $F1 = 72.4$. Source probes are trained after freezing the screening encoder, using a held-out embedding split. We report chance and balanced accuracy; conditional probes cover supported multi-view strata only, excluding single-source IMU strata.

Main Results

The main comparison asks whether support-gated source suppression improves screening utility while reducing recoverable source information. SourceGuard-Gait gives a favorable trade-off: it improves F1, recall, calibration, and worst-stratum F1 over shortcut-prone pooling. The over-aligned variant obtains the lowest probe score but drops recall to 74.6, showing why source recoverability must be interpreted together with screening evidence. The source-prior baseline receives no movement input and predicts $\hat{p} = P_{\text{train}}(Y = 1 | S = s)$ with threshold 0.5.

The strongest differences are in worst-stratum F1 and calibration, the metrics most directly tied to claim-bounded evaluation. CDAN, IRM, REx, and unconditional MMD improve over pooling but do not reach the same trade-off. Soft gating and rendering perturbation preserve the main conclusion, scalar temperature scaling improves ECE without changing thresholded decisions, and over-aligning lowers probe score at the cost of screening utility.

Stratified Screening Results

Table 3 reports held-out predictions by source. Reference sources are read through false-positive rate and sources with positives through recall. Public abnormal videos and small hemiplegic strata remain harder than rendered or wearable Parkinsonian sources, which is why condition-family reporting is part of the screening claim.

Source	Input	Subj. tr/val/te	Samples tr/val/te	Test neg.	Test pos.	Family	Rec./FPR	F1
WearGait-PD	IMU	51/7/15	1,284/184/392	166	226	Parkinsonian	89.4	89.6
CASIA-B	MV seq.	88/12/24	4,520/680/1,598	1,598	0	reference	4.0 FPR	–
Health&Gait	MV seq.	106/12/31	1,068/156/340	340	0	reference	5.3 FPR	–
Hemiplegic	MV seq.	27/5/9	122/21/41	21	20	hemiplegic	80.0	80.0
GAIT-IT/GAIT-IST	MV seq.	30/4/10	261/40/95	31	64	simulated	87.5	89.6
GAVD	MV seq.	100/12/31	1,264/187/423	130	293	abnormal video	82.6	85.3
CARE-PD	MV seq.	142/18/39	2,413/348/719	210	509	Parkinsonian	84.5	88.1
D-Gait	MV seq.	281/31/82	18,922/2,712/5,486	4,398	1,088	depression risk	87.2	77.5
Predictive total	IMU + MV seq.	825/101/241	29,854/4,328/9,094	6,894	2,200	public-source manifest	–	–

Table 3: Public-source manifest, subject-disjoint split, and source-stratified behavior. Derived units inherit the parent subject key; Rec./FPR reports recall for positive sources and false-positive rate for reference sources.

Method	F1	Rec.	Worst	ECE	Probe
Source-prior	53.1	67.5	28.4	31.6	–
Pooled	67.9	78.5	45.7	23.8	71.6
DANN	73.4	82.1	56.8	19.6	55.1
CDAN	75.1	83.0	58.7	18.1	54.9
MMD	73.7	78.2	55.9	18.2	53.6
GroupDRO	76.4	84.0	64.7	17.5	65.0
IRM	70.6	76.2	47.5	22.4	69.2
REx	72.1	80.7	52.3	20.7	67.5
+ sampler	72.2	82.8	56.4	20.5	64.8
+ gated GRL	77.4	85.2	66.1	15.6	56.5
+ gated MMD	79.2	85.7	69.8	14.2	53.6
soft gate	80.5	85.8	71.6	12.9	53.0
SourceGuard	80.8	86.1	72.4	12.7	52.8
Over-align	71.3	74.6	49.5	20.4	49.1
family gate	78.9	83.4	70.2	14.0	50.7
temp. scaled	80.8	86.1	72.4	10.9	52.8
render pert.	79.6	85.1	70.8	13.5	51.8
5-seed mean	80.5	85.8	71.9	12.9	53.1
SD	0.7	0.9	1.1	0.6	1.0

Table 4: Held-out screening trade-off and robustness checks. Probe is macro conditional source-probe balanced accuracy over supported MV strata; lower is better for ECE and probe.

Source-Probe Evaluation

Frozen probes measure source recoverability from learned embeddings. The macro conditional probe drops from 71.6 for pooled training to 52.8 for SourceGuard-Gait, with reductions in both supported MV routes. Scores remain above chance, so residual provenance is still measurable.

Ablation Study

The ablations separate three effects. Source-balanced sampling improves minority-source exposure but leaves substantial source recoverability. Gated GRL and MMD give the largest step in the screening/probe trade-off, and the soft gate reaches nearly the same operating point. The family-aware gate $q(m, y, c)$ lowers probe recovery slightly while removing useful cross-family screening evidence. Controls bound shortcuts, leakage, imbalance, and harder LOSO transfer ($F1 = 74.9$).

Discussion

In this paper, we present SourceGuard-Gait, a support-aware framework for abnormal-gait screening from heterogeneous public data. Specifically, we propose a claim-bounded invariance paradigm that uses a source–modality–label support matrix to determine where source suppression and distribution alignment are empirically justified. Moreover, we introduce frozen source probes to evaluate residual provenance information independently of the training-time source critic. Comprehensive experiments demonstrate that SourceGuard-Gait consistently improves screening performance, calibration, and worst-stratum robustness while reducing recoverable source information. In summary, we emphasize that heterogeneous gait learning is constrained not only by data scarcity, but also by incomplete cross-source support. SourceGuard-Gait provides a practical protocol for aligning robustness claims with the evidence available in each stratum. However, the current findings should be interpreted as public-source abnormal-gait screening rather than disease diagnosis, and count-based support cannot guarantee semantic or clinical comparability. Therefore, we anticipate that this framework will facilitate more transparent gait benchmarks and motivate future datasets with stronger cross-source overlap.

Limitations. The main limitations are data-level: screen-negative-only reference sources, small positive clinical strata, a binary label over distinct disease processes, and a count-based gate that cannot certify semantic or acquisition comparability. Calibration is manifest-specific; source probes remain above chance; stratum-level attribution is left for future work.

Ethical Statement

This work is for representation learning and screening support; clinical use requires prospective validation, institutional review, and access to public records under source terms. Generative AI tools assisted with manuscript editing, L^AT_EX formatting, and submission-compliance checks; the authors reviewed all resulting edits and remain fully responsible for the manuscript.

References

- Adeli, V.; Klabucar, I.; Rajabi, J.; Filtjens, B.; Mehraban, S.; Wang, D.; Seo, H.; Hoang, T.-H.; Do, M. N.; Muller, C.; et al. 2025. CARE-PD: A Multi-Site Anonymized Clinical Dataset for Parkinson's Disease Gait Assessment. In *Advances in Neural Information Processing Systems*, volume 38. Datasets and Benchmarks Track.
- Anderson, A. J.; Eguren, D.; Gonzalez, M. A.; Caiola, M.; Khan, N.; Watkinson, S.; Zuccaroli, I.; Hirczy, S. S.; Zabetian, C. P.; Mills, K.; Moukheiber, E.; Moro-Velazquez, L.; Dehak, N.; Motley, C.; Muir, B. C.; Butala, A.; and Kontson, K. 2026. WearGait-PD: An Open-Access Wearables Dataset for Gait in Parkinson's Disease and Age-Matched Controls. *Scientific Data*, 13(1): 440.
- Arjovsky, M.; Bottou, L.; Gulrajani, I.; and Lopez-Paz, D. 2019. Invariant Risk Minimization. arXiv:1907.02893.
- Celik, Y.; Stuart, S.; Woo, W. L.; Sejdic, E.; and Godfrey, A. 2022. Multi-Modal Gait: A Wearable, Algorithm and Data Fusion Approach for Clinical and Free-Living Assessment. *Information Fusion*, 78: 57–70.
- Chao, H.; He, Y.; Zhang, J.; and Feng, J. 2019. GaitSet: Regarding Gait as a Set for Cross-View Gait Recognition. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 33, 8126–8133.
- Cui, Y.; and Kang, Y. 2023. Multi-Modal Gait Recognition via Effective Spatial-Temporal Feature Fusion. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 17949–17957.
- DeGrave, A. J.; Janizek, J. D.; and Lee, S.-I. 2021. AI for Radiographic COVID-19 Detection Selects Shortcuts over Signal. *Nature Machine Intelligence*, 3: 610–619.
- Duan, B.; Wan, X.; and Zhao, X. 2024. FSGait: Fine-Grained Self-Supervised Gait Abnormality Detection. In *Computer Vision – ACCV 2024*, Lecture Notes in Computer Science, 313–329. Springer Nature Singapore.
- Fan, C.; Peng, Y.; Cao, C.; Liu, X.; Hou, S.; Chi, J.; Huang, Y.; Li, Q.; and He, Z. 2020. GaitPart: Temporal Part-Based Model for Gait Recognition. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 14225–14233.
- Ganin, Y.; Ustinova, E.; Ajakan, H.; Germain, P.; Larochelle, H.; Laviolette, F.; Marchand, M.; and Lempitsky, V. 2016. Domain-Adversarial Training of Neural Networks. *Journal of Machine Learning Research*, 17(59): 1–35.
- Geirhos, R.; Jacobsen, J.-H.; Michaelis, C.; Zemel, R.; Brendel, W.; Bethge, M.; and Wichmann, F. A. 2020. Shortcut Learning in Deep Neural Networks. *Nature Machine Intelligence*, 2: 665–673.
- Gretton, A.; Borgwardt, K. M.; Rasch, M. J.; Scholkopf, B.; and Smola, A. 2012. A Kernel Two-Sample Test. *Journal of Machine Learning Research*, 13(25): 723–773.
- Gu, X.; Guo, Y.; Deligianni, F.; Lo, B.; and Yang, G.-Z. 2021. Cross-Subject and Cross-Modal Transfer for Generalized Abnormal Gait Pattern Recognition. *IEEE Transactions on Neural Networks and Learning Systems*, 32(2): 546–560.
- Gulrajani, I.; and Lopez-Paz, D. 2021. In Search of Lost Domain Generalization. In *International Conference on Learning Representations*.
- Guo, C.; Pleiss, G.; Sun, Y.; and Weinberger, K. Q. 2017. On Calibration of Modern Neural Networks. In *Proceedings of the 34th International Conference on Machine Learning*, volume 70 of *Proceedings of Machine Learning Research*, 1321–1330. PMLR.
- Guo, X.; Zhu, Z.; Yang, T.; Lin, B.; Huang, J.; Deng, J.; Huang, G.; and Zhou, J. 2025. Gait Recognition in the Wild: A Large-Scale Benchmark and NAS-Based Baseline. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 47(6): 4535–4552.
- Guo, Y.; Gu, X.; and Yang, G.-Z. 2021. MCDGD: Multi-Source Unsupervised Domain Adaptation for Abnormal Human Gait Detection. *IEEE Journal of Biomedical and Health Informatics*, 25(10): 4017–4028.
- Han, J.; and Bhanu, B. 2006. Individual Recognition Using Gait Energy Image. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 28(2): 316–322.
- IAPR Technical Committee 4. n.d. Gait Datasets. <https://iapr-tc4.org/gait-datasets/>; accessed: 2026-07-13.
- Institute of Automation, Chinese Academy of Sciences. 2016. CASIA Gait Databases. Official database index page, published online 2016-10-26; https://english.ia.cas.cn/db/201610/t20161026_169403.html; accessed: 2026-07-13.
- Instituto de Telecomunicacoes. n.d.a. GAIT-IST Dataset. <http://www.img.lx.it.pt/GAIT-IST/>; accessed: 2026-07-13.
- Instituto de Telecomunicacoes. n.d.b. GAIT-IT Dataset. <http://www.img.lx.it.pt/GAIT-IT/>; accessed: 2026-07-13.
- Jun, K.; Lee, S.; Lee, D.-W.; and Kim, M. S. 2021. Deep Learning-Based Multimodal Abnormal Gait Classification Using a 3D Skeleton and Plantar Foot Pressure. *IEEE Access*, 9: 161576–161589.
- Koh, P. W.; Sagawa, S.; Marklund, H.; Xie, S. M.; Zhang, M.; Balsubramani, A.; Hu, W.; Yasunaga, M.; Phillips, R. L.; Gao, I.; et al. 2021. WILDS: A Benchmark of in-the-Wild Distribution Shifts. In *Proceedings of the 38th International Conference on Machine Learning*, volume 139 of *Proceedings of Machine Learning Research*, 5637–5664. PMLR.
- Kontson, K.; Butala, A.; Muir, B.; Anderson, A.; Caiola, M.; Eguren, D.; Gonzalez, M.; Motley, S.; Mills, K.; Moro Velazquez, L.; Dehak, N.; and Moukheiber, E. 2024. Wearables for Gait in Parkinson's Disease and Age-Matched Controls (WearGait-PD). Synapse. Dataset; accessed: 2026-07-13.
- Krueger, D.; Caballero, E.; Jacobsen, J.-H.; Zhang, A.; Binias, J.; Zhang, D.; Le Priol, R.; and Courville, A. 2021. Out-of-Distribution Generalization via Risk Extrapolation (REX). In *Proceedings of the 38th International Conference on Machine Learning*, volume 139 of *Proceedings of Machine Learning Research*, 5815–5826. PMLR.
- Lin, T.-Y.; Goyal, P.; Girshick, R.; He, K.; and Dollar, P. 2017. Focal Loss for Dense Object Detection. In *Proceedings of the IEEE International Conference on Computer Vision*, 2980–2988.

- Liu, X.; Li, Q.; Hou, S.; Ren, M.; Hu, X.; and Huang, Y. 2024. Depression Risk Recognition Based on Gait: A Benchmark. *Neurocomputing*, 596: 128045.
- Long, M.; Cao, Z.; Wang, J.; and Jordan, M. I. 2018. Conditional Adversarial Domain Adaptation. In Bengio, S.; Wallach, H.; Larochelle, H.; Grauman, K.; Cesa-Bianchi, N.; and Garnett, R., eds., *Advances in Neural Information Processing Systems*, volume 31, 1640–1650. Curran Associates, Inc.
- Loureiro, J.; and Correia, P. L. 2020. Using a Skeleton Gait Energy Image for Pathological Gait Classification. In *Proceedings of the 15th IEEE International Conference on Automatic Face and Gesture Recognition*, 503–507.
- Naz, A.; Prasad, P. S.; Mccall, S.; Leung, C. C.; Ochi, I.; Gong, L.; and Yu, M. 2025. Privacy-Preserving Abnormal Gait Detection Using Computer Vision and Machine Learning. *EAI Endorsed Transactions on Pervasive Health and Technology*, 11.
- Ong Ly, C.; Unnikrishnan, B.; Tadic, T.; Patel, T.; Duhamel, J.; Kandel, S.; Moayedi, Y.; Brudno, M.; Hope, A.; Ross, H.; and McIntosh, C. 2024. Shortcut Learning in Medical AI Hinders Generalization: Method for Estimating AI Model Generalization without External Data. *npj Digital Medicine*, 7: 124.
- Ranjan, R.; Ahmedt-Aristizabal, D.; Ali Armin, M.; and Kim, J. 2025a. Computer Vision for Clinical Gait Analysis: A Gait Abnormality Video Dataset. *IEEE Access*, 13: 45321–45339.
- Ranjan, R.; Ahmedt-Aristizabal, D.; Ali Armin, M.; and Kim, J. 2025b. GAVD: Gait Abnormality in Video Dataset. GitHub repository. <https://github.com/Rahmyyy/GAVD>; accessed: 2026-07-13.
- Sagawa, S.; Koh, P. W.; Hashimoto, T. B.; and Liang, P. 2020. Distributionally Robust Neural Networks for Group Shifts: On the Importance of Regularization for Worst-Case Generalization. In *International Conference on Learning Representations*.
- Schroff, F.; Kalenichenko, D.; and Philbin, J. 2015. FaceNet: A Unified Embedding for Face Recognition and Clustering. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, 815–823.
- Sun, B.; and Saenko, K. 2016. Deep CORAL: Correlation Alignment for Deep Domain Adaptation. In *European Conference on Computer Vision Workshops*, 443–450.
- Teepe, T.; Gilg, J.; Herzog, F.; Hormann, S.; and Rigoll, G. 2022. Towards a Deeper Understanding of Skeleton-Based Gait Recognition. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops*, 1568–1576.
- Wang, Z.; Hou, S.; Zhang, M.; Liu, X.; Cao, C.; Huang, Y.; Li, P.; and Xu, S. 2024. QAGait: Revisit Gait Recognition from a Quality Perspective. *Proceedings of the AAAI Conference on Artificial Intelligence*, 38(6): 5785–5793.
- Yang, Y.; Yang, X.; Sakamoto, T.; Fioranelli, F.; Li, B.; and Lang, Y. 2022. Unsupervised Domain Adaptation for Disguised-Gait-Based Person Identification on Micro-Doppler Signatures. *IEEE Transactions on Circuits and Systems for Video Technology*, 32(9): 6448–6460.
- Yu, S.; Tan, D.; and Tan, T. 2006. A Framework for Evaluating the Effect of View Angle, Clothing and Carrying Condition on Gait Recognition. In *Proceedings of the 18th International Conference on Pattern Recognition*, volume 4, 441–444.
- Zafra-Palma, J.; Marin-Jimenez, N.; Castro-Pinero, J.; Cuenca-Garcia, M.; Munoz-Salinas, R.; and Marin-Jimenez, M. J. 2025. Health & Gait: A Dataset for Gait-Based Analysis. *Scientific Data*, 12(1): 44.
- Zhou, K.; Liu, Z.; Qiao, Y.; Xiang, T.; and Loy, C. C. 2023. Domain Generalization: A Survey. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 45(4): 4396–4415.
- Zhou, K.; Yang, Y.; Qiao, Y.; and Xiang, T. 2021. Domain Generalization with MixStyle. In *International Conference on Learning Representations*.